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Comparison of Deconvolution Techniques for Spatially Resolved Transcriptomics Data

Introduction

Spatially Resolved Transcriptomics (ST) is a molecular profiling technique that allows insight into tumor heterogeneity and the composition of tumor microenvironment data [1]. Current ST techniques typically provide a multi-cell resolution with cell mixtures for small spatial areas ('spots'). Computational deconvolution methods can then be employed to approximate the expression and type of the involved cells per spot. While the algorithms STdeconvolve (SD) [2] and SPOTlight (SL) [3] offer functions for spot-wise ST deconvolution, the Seurat [4] pipeline uses an integration approach (SI) to predict the cell composition for each ST spot.

Methods

Three computational deconvolution and integration algorithms for ST data were compared on a set of eight original human 10X Visium retinoblastoma samples (Figure 1). Matched single-cell RNA-seq (scRNA) data with detailed annotations was available as reference dataset. Deconvolutions for different topic numbers and levels of details for the reference annotation were created and the run time of the algorithms was tested with basic R functions.

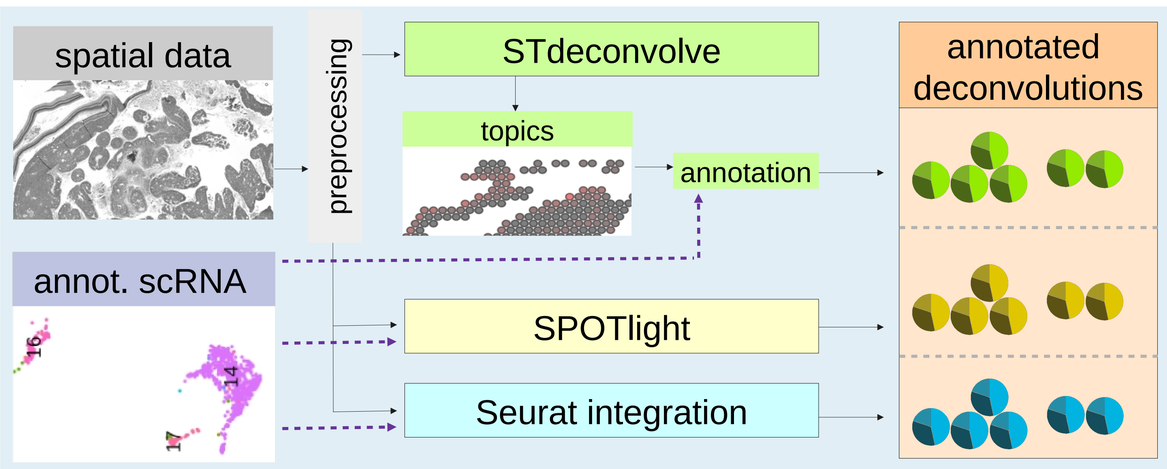

Figure 1: Analysis workflow. After a preprocessing step with SpaceRanger 1.3.0, spot-wise deconvolutions were calculated for the ST data and annotated based on the matched scRNA data.

Results

The Seurat integration approach and SPOTlight both indicated a high proportion of tumor cells (general tumor or tumor cones) in most of the ST clusters. For most STdeconvolve cell types, correlation results were high for the different tumor types as well. The Seurat integration indicated a notable presence of glia cells in clusters 1 and 6, and bipolar cells in cluster 6 (Figure 2), which matched the marker-based expert annotation of the dataset.

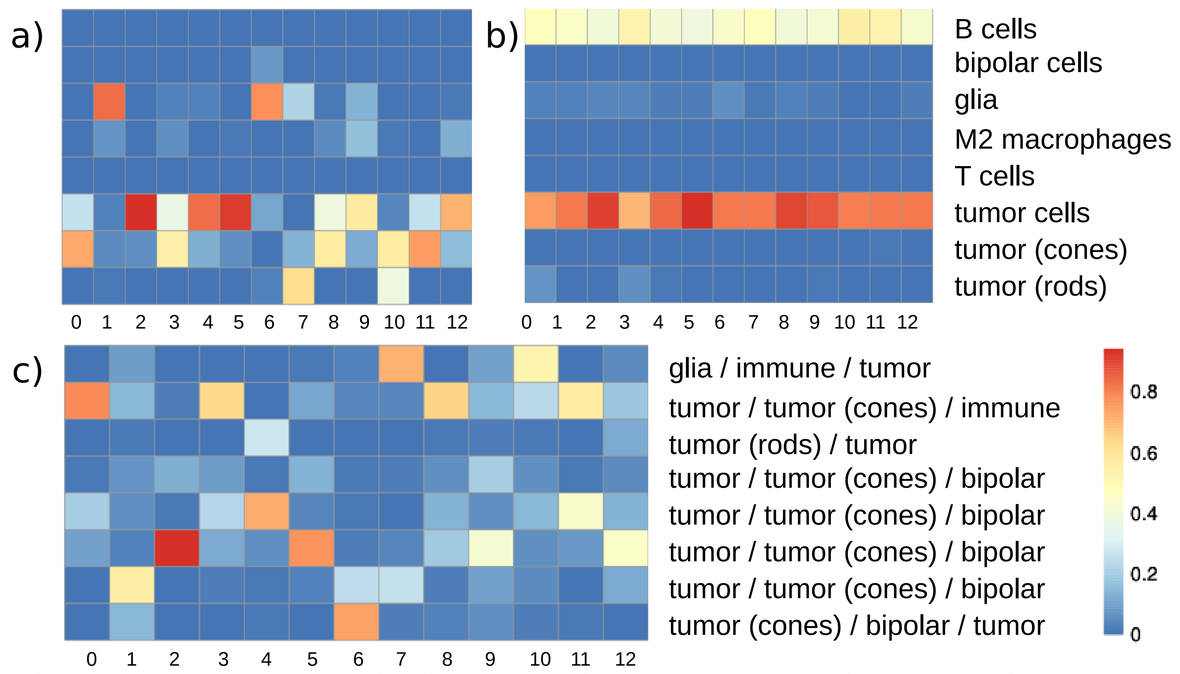

Figure 2: Deconvolution results per ST cluster. a) Seurat integration (prediction scores), b) SPOTlight (deconvolution matrix), c) STdeconvolve (correlation to reference cell types).

In contrast, SPOTlight mainly identified tumor cells and B cells in the dataset. STdeconvolve's cell types usually showed correlations of 0.5 or higher with multiple reference cell types, with a general trend for high tumor and tumor cones proportions, and bipolar cells. This pattern was similar for all eight samples.

Median run times in minutes were comparable between algorithms (SL=6.06, SI=8.76, SD=13.8). All three algorithms are integrated in the R/Bioconductor environment and include detailled documentation.

Discussion & Conclusion

Deconvoluted cell types and marker-based annotation of ST clusters by domain experts varied depending on the cell proportions and composition of the underlying tissue, with higher concordance for transcriptionally distinct cell types, and cell types with a high frequency in both ST and scRNA data. The Seurat integration generally yielded the most accurate results for the eight retinoblastoma samples. Further research is required to achieve more consistent and reliable results for less frequent or more similar cell types; an improved resolution of ST data is desirable.

References

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